



DOW™ Wire & Cable HFDG-4202

Crosslinkable Power Cable Insulation Compound

Overview

HFDG-4202 is a low density, water-tree retardant, crosslinkable, unfilled polyethylene insulation compound designed for use in power cable applications of 10 kV or less.

Specifications

Cables insulated with HFDG-4202, using sound commercial fabrication manufacturing practice, would be expected to meet the latest editions of the following specifications and regulations:

- IEC: 60502
- GB/T 12706

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density ¹ (73°F (23°C))	0.920 g/cm ³	0.920 g/cm ³	ASTM D792
Degree of Crosslinking - Extractables	< 20 %	< 20 %	ASTM D2765A
Mechanical	Nominal Value (English)	Nominal Value (SI)	Test Method
Tensile Strength ²	2800 psi	19.3 MPa	ASTM D638
Tensile Elongation ²	500 %	500 %	ASTM D638
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Hot Creep (392°F (200°C))	< 100 %	< 100 %	ICEA T-28-562
Hot Set (392°F (200°C))	< 10 %	< 10 %	ICEA T-28-562
Electrical	Nominal Value (English)	Nominal Value (SI)	Test Method
Volume Resistivity ³ (73°F (23°C))	> 1.0E+15 ohms·cm	> 1.0E+15 ohms·cm	ASTM D257
Dielectric Strength ⁴			ASTM D149
0.125 in (3.18 mm), Method A (Short-Time)	660 V/mil	26 kV/mm	
0.125 in (3.18 mm), Method B (Step-by-Step)	580 V/mil	23 kV/mm	
Dielectric Constant ⁵ (73°F (23°C), 60 Hz)	2.30	2.30	ASTM D150
Dissipation Factor ⁵ (73°F (23°C), 60 Hz)	5.0E-4	5.0E-4	ASTM D150

Additional Information

Nominal property values representing tests on molded stress-relieved slabs. Cure times were 15 minutes at 175°C.

Storage

The environment or conditions of storage greatly influences the recommended storage time. Storage under extreme conditions may affect the quality, processing, or performance of the product. Storage should be in accordance with good manufacturing practices. The recommended storage conditions, in the original unopened packages, are dry conditions with temperatures between 50°F and 104°F (10°C and 40°C). When stored between 50°F and 86°F (10°C and 30°C), the product may be used by the customer for up to one year from the date of sale or two years from the date of manufacture, whichever comes first. The recommended maximum storage time is 1 year at 104°F (40°C). It is recommended that the practice of using the product on a first-in / first-out basis be established.

Packaging

HFDG-4202 can be delivered in 1300lb/590 kg top unloading octabins.

Extrusion Notes

HFDG-4202 provides excellent performance and outstanding output rates over a range of extrusion conditions. For optimum results, melt extrusion temperatures in the range of 115°C to 140°C (240°F to 285°F) are recommended, although higher melt temperatures may be possible on certain extrusion lines with due care. Specific processing recommendations can only be made when information about the application and actual extrusion and processing equipment types are known.

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ ASTM D792/ISO 1183

² ASTM D638/IEC 60811-1-1

³ ASTM D257/IEC 60093

⁴ ASTM D149/IEC 60243-1

⁵ ASTM D150/IEC 60250

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